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For each of the sections below, your reported test accuracy should approximately match the accuracy reported on Kaggle.

Perceptron

For MNIST: lr is searched uniformly in [0.5, 0.8]; epochs is searched one by one between 5 and 20; decay_rate is searched uniformly in [1e-4, 1e-1].
For RICE: lr test range: [0.1, 0.3, 0.5, 0.7, 1.0], grid search; epoch test range: [5, 10, 15, 20], grid search
After testing different learning rate decay values, it is found that the accuracy is lower than the current one when the decay rate is 0.0, which proves that learning rate decay is slightly useful.

RICE DATASET

Optimal hyperparameters:	lr = 0.5 n_epochs = 10 decay_rate = 0.0007098068775694665
Training accuracy:	99.468426
Validation accuracy:	99.477591
Test accuracy:	99.422601

Fashion-MNIST DATASET

Optimal hyperparameters:	lr = 0.5883478742594644 n_epochs = 11 decay_rate = 0.0007098068775694665
Training accuracy:	83.286000
Validation accuracy:	81.090000
Test accuracy:	80.380000

SVM

For MNIST: lr is searched uniformly in [1e-3,1.0]; epochs is searched one by one between 5 and 20; reg_const is searched uniformly in [1e-7,5.0]; batch_size is selected one by one between 64, 128, 256; decay_rate is searched uniformly in [1e-3,1e-1]
For RICE: lr is searched uniformly in [1e-3,1.0]; epochs is searched one by one between 5 and 200; reg_const is searched uniformly in [1e-4,10.0]
When mini-batch is not used and decay_rate is 0, the accuracy decreases. Therefore, these two methods are useful.

RICE DATASET

Optimal hyperparameters:	lr = 0.057068171943404314 n_epochs = 172 reg_const = 0.0003122299278112629 batch_size = 128 decay_rate = 0.002091426077065719
Training accuracy:	99.926679
Validation accuracy:	99.862524
Test accuracy:	99.890019

Fashion-MNIST DATASET

Optimal hyperparameters:	lr = 0.035983073221155626 n_epochs = 11 reg_const = 3.362309019476433e-05 batch_size = 128 decay_rate = 0.002091426077065719
Training accuracy:	83.472000
Validation accuracy:	82.290000
Test accuracy:	80.630000

Softmax

For MNIST: lr is searched uniformly in [1e-3,1.0]; epochs is searched one by one between 5 and 20; reg_const is searched uniformly in [1e-4,0.5]; batch_size is selected one by one between 128, 256, 512; decay_rate is searched uniformly in [1e-5,1e-1]; temperature is searched uniformly in [0.5,2.0]
For RICE: lr test range: [0.1, 0.3, 0.5, 0.7, 1.0], grid search; epoch test range: [5, 10, 15, 20], grid search; reg_const test range: [0.01, 0.03, 0.05, 0.1, 0.15] , Grid Search
When mini-batch, temperature, and decay_rate are not used, the accuracy decreases. Therefore, these three methods are useful.

RICE DATASET

Optimal hyperparameters:	lr = 0.5 n_epochs = 10 reg_const = 0.05
Training accuracy:	98.625241
Validation accuracy:	98.680231
Test accuracy:	98.542755

Fashion-MNIST DATASET

Optimal hyperparameters:	lr = 0.010872296945794026 n_epochs = 12 reg_const = 0.003457774444224259 batch_size = 256 decay_rate = 0.012949184782211806 temperature = 0.7008162465836012
Training accuracy:	84.794000
Validation accuracy:	82.740000
Test accuracy:	81.920000

Logistic

lr test range: [0.1, 0.3, 0.5, 0.7, 1.0], grid search; epoch test range: [5, 10, 15, 20], grid search; threshold test range: [0.3, 0.4, 0.5, 0.6, 0.7], grid search

RICE DATASET

Optimal hyperparameters:	learning_rate = 0.3 n_epochs = 15 threshold = 0.4
Training accuracy:	97.772890

Validation accuracy:	98.020346
Test accuracy:	97.745395